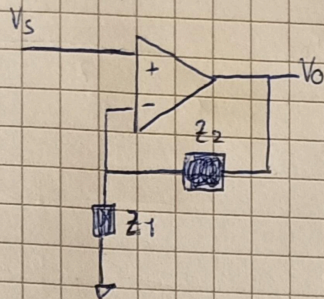


Block diagrams

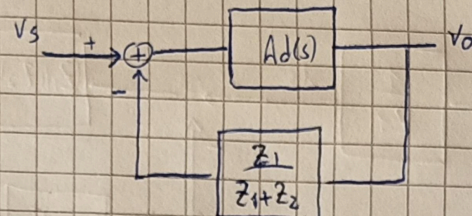
Non-inverting configuration

$$A_F = A_V = 1 + \frac{z_2}{z_1}$$

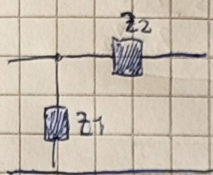


Same block diagram using direct loop and feedback theory

→ direct loop inspection: $T = -\frac{x_r}{x_s} = A_d(s) \cdot \frac{\beta}{z_1 + z_2}$



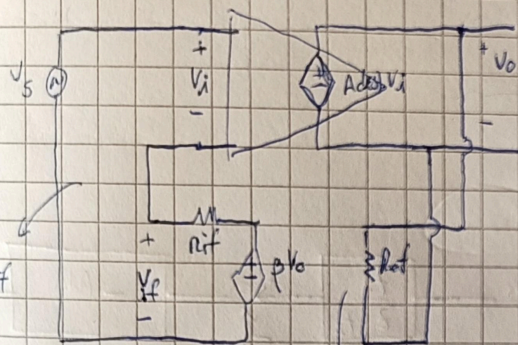
→ feedback theory: voltage mixing, voltage sampling



$$R_{if} = z_1 \parallel z_2$$

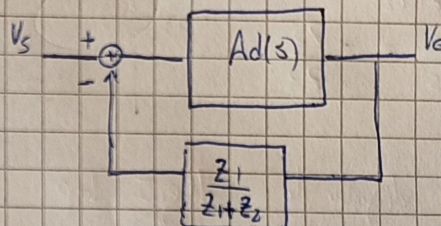
$$\beta = \frac{z_1}{z_1 + z_2}$$

$$R_{of} = z_1 + z_2$$



will not consider R_{if} because $R_{if} \rightarrow \infty$

will not consider R_{of} because $R_{of} \rightarrow 0$



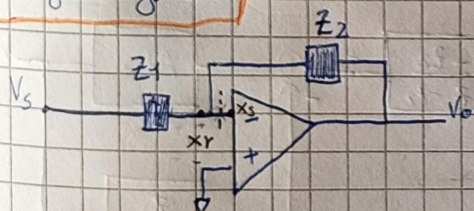
$$V_o = A_d(s) \left[V_s - V_o \frac{z_1}{z_1 + z_2} \right]$$

$$V_o \left[1 + A_d(s) \frac{z_1}{z_1 + z_2} \right] = A_d(s) V_s$$

$$A_V = \frac{V_o}{V_s} = \frac{A_d(s)}{1 + A_d(s) \frac{z_1}{z_1 + z_2}} = A_F = \frac{A_d}{1 + \beta A_d}$$

ideally $A_d(s) \rightarrow \infty \rightarrow A_V = A_F = 1 + \frac{z_2}{z_1}$

Inverting configuration

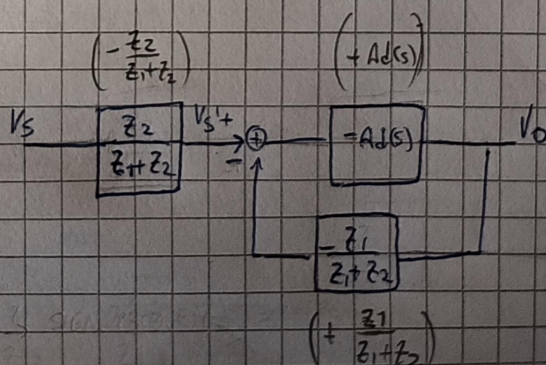


different block diagrams for direct loop and feedback theory analysis

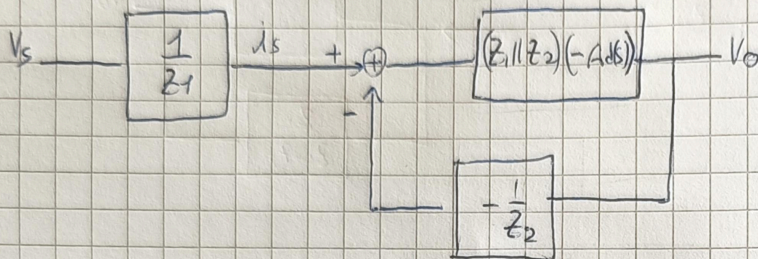
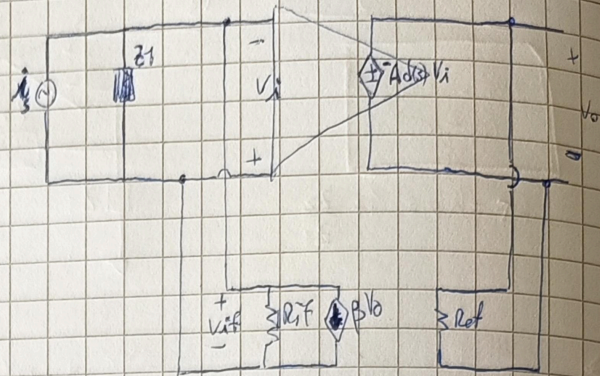
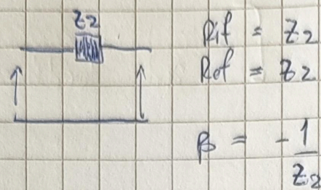
→ direct loop inspection: $T = -\frac{x_r}{x_s} = -A_d(s) \cdot \frac{\beta}{z_1 + z_2}$

$$V_o = -A_d(s) \left[V_s + V_o \frac{z_1}{z_1 + z_2} \right]$$

$$\frac{V_o}{V_s} = \frac{-A_d(s)}{1 + A_d(s) \frac{z_1}{z_1 + z_2}} \xrightarrow{A_d(s) \rightarrow \infty} -\frac{z_1 + z_2}{z_1} \rightarrow \frac{V_o}{V_s} = -\frac{z_2}{z_1}$$



→ feedback theory: current mixing, voltage sampling

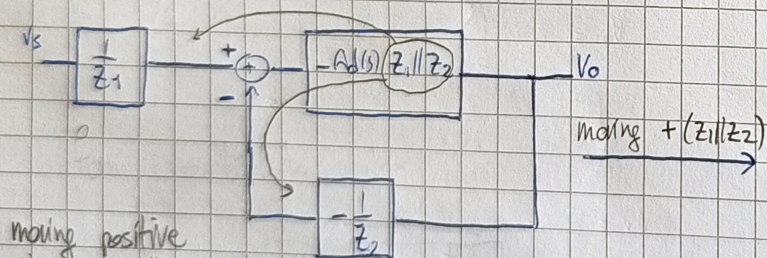


$$V_o = \left[\frac{Z_1 Z_2}{Z_1 + Z_2} (-Ad(s)) \right] \left(i_s + \frac{V_o}{Z_2} \right) \rightarrow V_o \left[1 + \frac{Z_1}{Z_1 + Z_2} Ad(s) \right] = -Ad(s) i_s \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$\frac{V_o}{i_s} = \frac{\frac{Z_1 Z_2}{Z_1 + Z_2} (-Ad(s))}{1 + \frac{Z_1}{Z_1 + Z_2} Ad(s)} \xrightarrow{Ad(s) \rightarrow \infty} \frac{V_o}{i_s} = -\frac{Z_1 Z_2}{Z_1 + Z_2} \cdot \frac{Z_1 + Z_2}{Z_1} = -Z_2$$

$$\frac{V_o}{V_s} = \frac{V_o}{i_s} \frac{i_s}{V_s} = -\frac{Z_2}{Z_1}$$

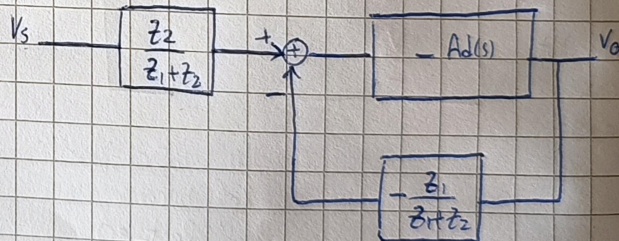
→ From feedback diagram to direct loop



moving positive things nothing changes

moving $+(Z_1 || Z_2)$

moving $-(Z_1 || Z_2)$



↑ equivalent

